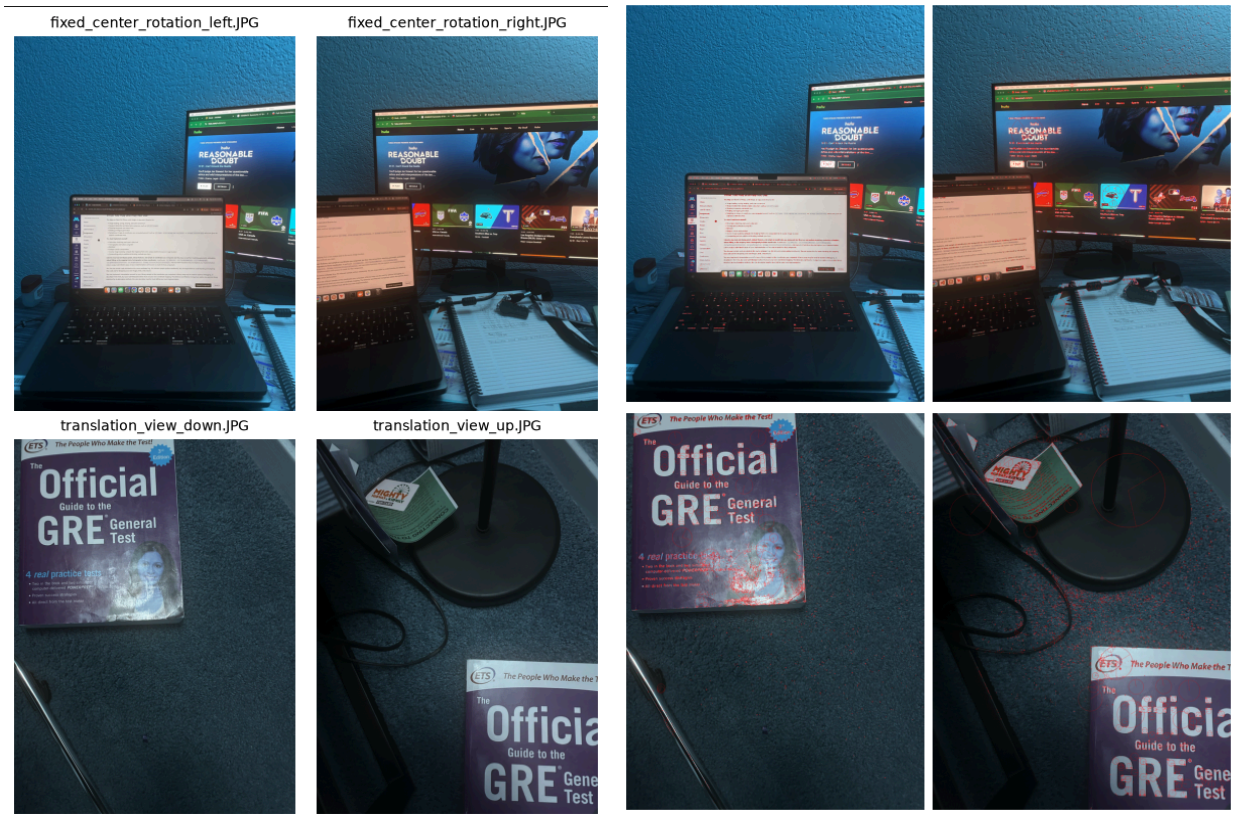
Assignment item	Points	Location in report	Extra credit?
Report Quality	5	-	No
Image Capture and Feature Extraction	10	Section 2, p. 2	No
Descriptor Matching	10	Section 3, p. 2-3	No
Homography Estimation	10	Section 4, p. 3	No
Basic Robust Estimation with RANSAC	20	Section 5, p. 3-4	No
Warping and Compositing	15	Section 6, pp. 4–6	No
Confidence-Based RANSAC Stopping	10	Section 7, p. 6	No
RANSAC Parameter Analysis	10	Section 8, p. 6-7	No
Detector and Matcher Comparison	15	Section 9, p. 7-8	Partially
Interpolation Comparison	5	Section 10 p. 8	Yes
Total attempted before extra-credit conversion	**110**		
Maximum possible score after conversion	**105**		

Panoramic Image Stitching

In this project, I implemented a panoramic image stitcher from scratch. The system takes in two or more overlapping images, detects and matches features, creates a homography, and warps them into a common coordinate frame for a panoramic image. I created two image pair sets. The first is two images of my desk, to test rotation, with my laptop, monitor, cables, and notebook. The second is two images of my GRE test prep book, to test translation. In this paper I will walk you through the various algorithms needed in the panoramic image stitching process.

I captured the dataset using my IPhone 14 camera and items that were on my desk while I was doing this assignment. A homography was appropriate for these two image sets because there is very minimal depth variation of the camera, the set of my laptop just has rotation, while the set with my GRE prep book has pure translation. For each image there were 8000 keypoints as I set the nfeatures parameter for SIFT to be 8000, I made no other parameter changes. The image on the left is the dataset while on the right is the keypoint visualization, all the keypoints are drawn on the image as red dots.



I used a lowe's ratio test threshold of 0.75, which I determined through trial and error of testing multiple different values for the best fitting image stitch. There were 8000 candidate descriptors before the lowe's ratio test for both image pairs and after there were 1145 for the laptop image and 497 for the gre prep book image. I used k=2 in my K nearest neighbors which

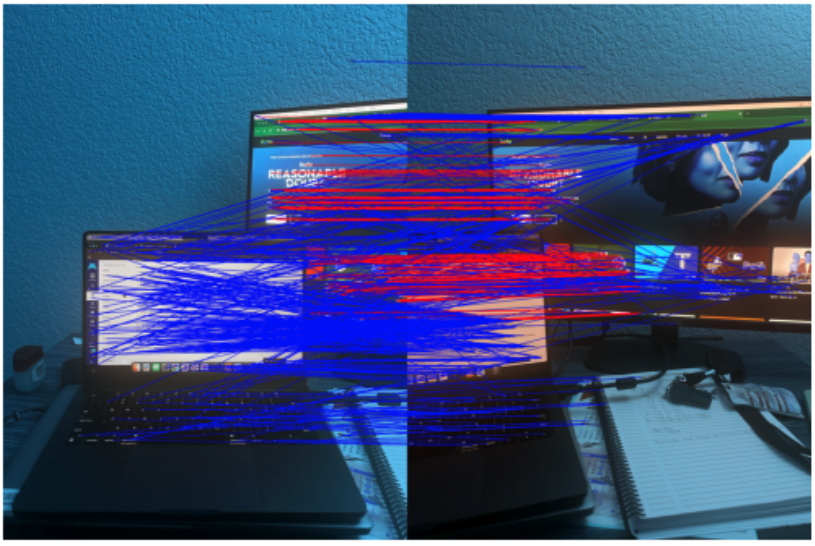
means 16000 actual points. On the left is the one match line visualization before and on the right is after the lowe's ratio test.



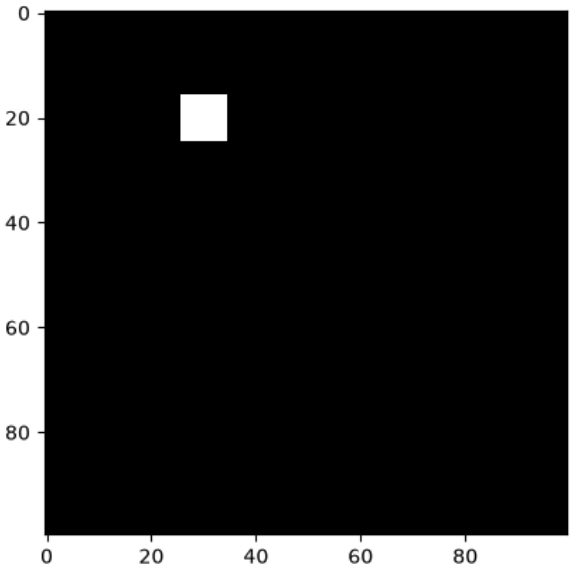


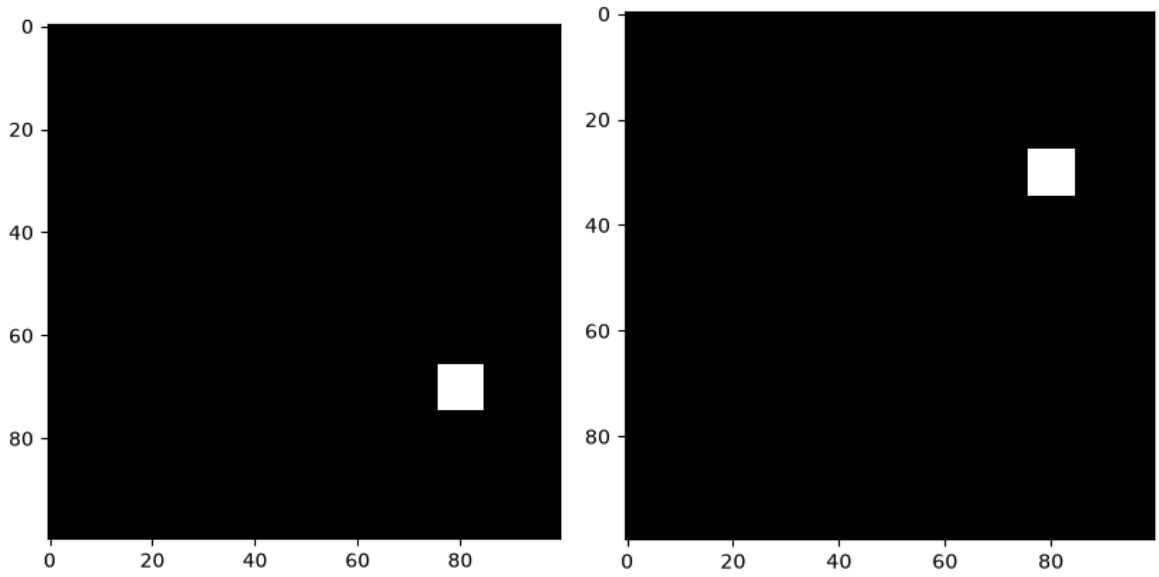
The known homography that I chose was [[0.9, 0.1, 40.0], [-0.05, 1.1, 20.0], [0.0003, -0.0002, 1.0]]. In the first test I chose 4 points and the second I chose 8, my goal was to make the points non collinear so I focused on making quadrilaterals when choosing my points. The reprojection error for both test was approximately 0 as the one with 4 points was 6.8e-06 and the one with 8 points was 1.15e-05. In my ransac implementation I prevented degenerate 4 point samples by checking if the homography was not invertible and discarding if true.

My initial inlier threshold was 3 and iteration budget was 1000, determined through a little trial and error to see where the inlier count plateaus, though I later implemented an adaptive stopping and parameter analysis. For the laptop image set before the refit there was an inlier count of 377, total match count of 1145, inlier ratio of 0.3292, and mean inlier reprojection error of 1.202. After the refit there was an inlier count of 399, total match count of 1145, inlier ratio of 0.3485, and mean inlier reprojection error of 0.86. For the gre prep book image set before the refit there was an inlier count of 175 total match count of 497, inlier ratio of 0.35, and mean inlier reprojection error of 1.31. After the refit there was an inlier count of 187, total match count of 497, inlier ratio of 0.376, and mean inlier reprojection error of 1.25. I did the refitting by creating a list that had all the inlier points from the best fit and finding a homography using that set of points. There is slight variation in the inlier count now that the fit is based on many more points rather than just 4 which can suffer from noise. The red lines are inliers, while the blue lines are outliers.



I sampled the source image by using inverse warping with a bilinear interpolation strategy. I looped through every coordinate on the target canvas and looked up the source image value. Here the first image on the top row is the original synthetic test image. On the bottom row are the warped images, the left is a pure translation of 50 pixels to the right and down, on the right is a 90 degree rotation of the image. We can see that the identifiable landmark, the white square, moved as expected as in the bottom left it moved 50 pixels to the right and down, while on the right image it rotated 90 degrees clockwise.

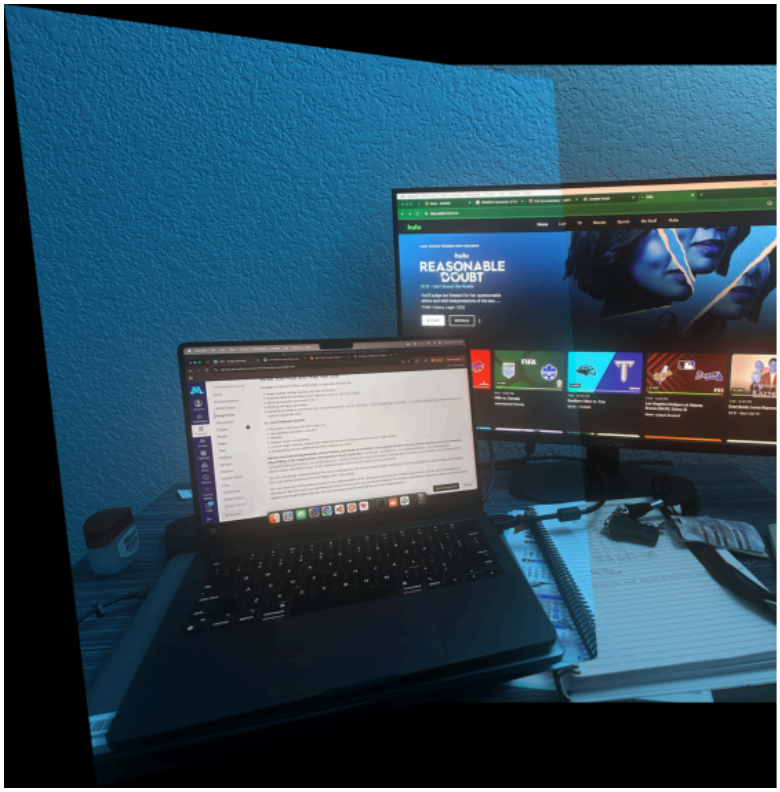




For the final panoramic images the canvas dimension of the laptop set was a height of 4959 and a width of 4875, fo the gre test book the canvas dimensions were height of 8361 and width of 5246. For both panoramic images the reference image is the one on the right.

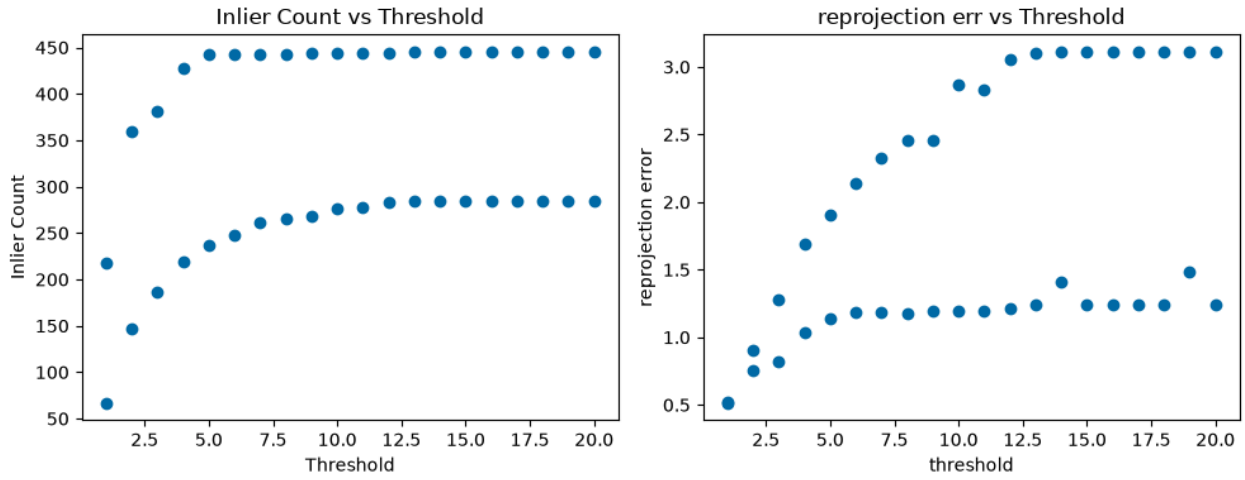


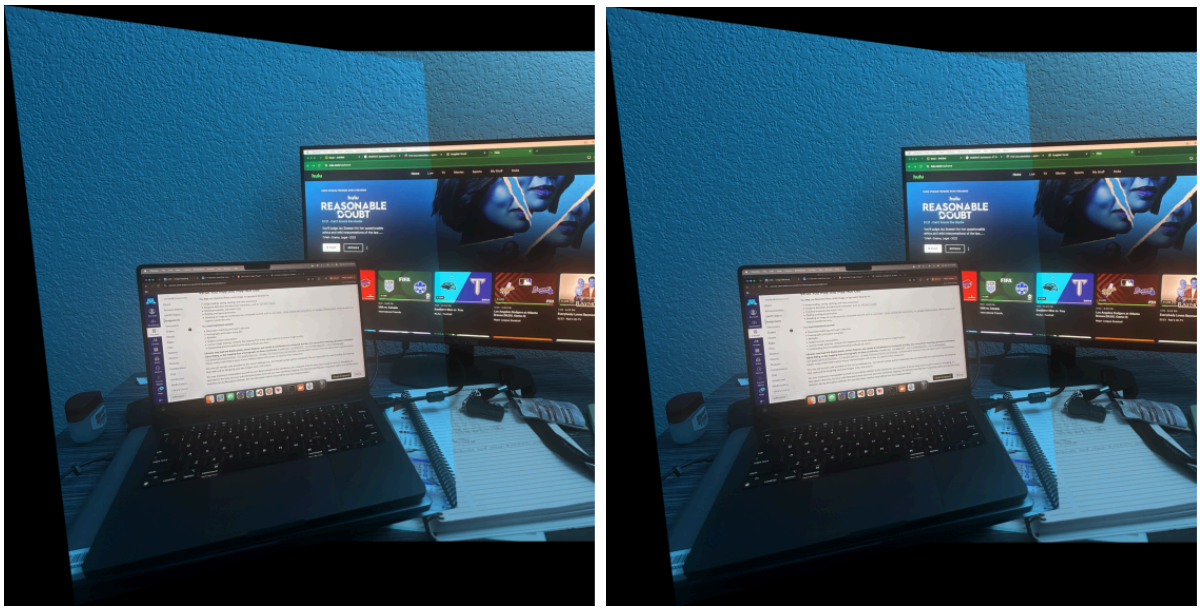




In my Confidence-Base RANSAC Stopping I used a confidence target of 99%. My formula was N = log(1-p)/log(1-w^s), where N is total iterations, p confidence target, w is inlier ratio, and s is how many points you are fitting. The confidence based RANSAC for the laptop image set takes about 374 iterations before stopping and finds 428 inliers, while the original RANSAC used 1000 iterations and found 399 inliers. For my test prep book image, the confidence based one used 335 iterations to find 183 inliers, while the original used 1000 iterations to find 187 inliers. As we can see the confidence based RANSAC is able to determine when to stop fitting much earlier with very similar inlier counts, the slight variance is due to the random sampling that RANSAC does.

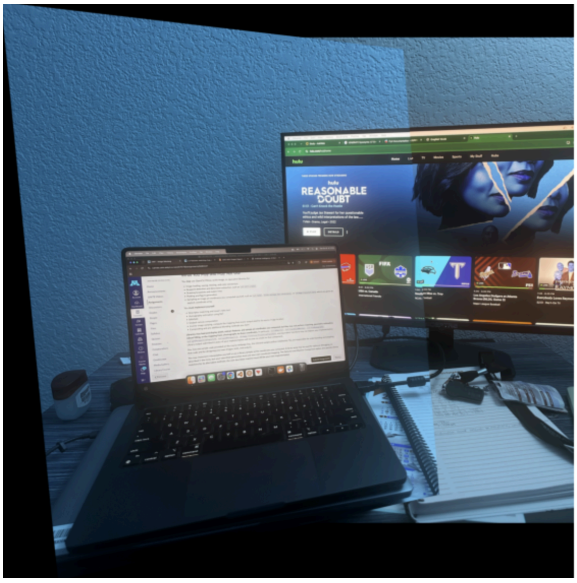
On the scatter charts below in the left chart the top line is the laptop image, while on the right chart the bottom line is the laptop image. We can see that both for both inlier count a reprojection error plateau at a certain threshold for each image, for the laptops set it is about 5 while for the test prep book set it is 11-12. This makes sense as the laptop set had significantly more key points so it can afford to be more stingy on its inliers while the book set had less so it must accept what it can get. The left laptop image is at a threshold of 1 while the right is at a threshold of 5 where it plateaus. We can see that at the lower threshold of the laptop image the stitch is at slightly more of an angle. This causes some of the text at the top of the monitor screen by the tabs to be slightly less aligned and at more of an angle than when compared to higher threshold.





When comparing SIFT, ORB, and AKAZE though they have some similarities they produce wildly different results. All three find 8000 keypoints as that is the threshold that I have set. SIFT finds 1145 matches while ORB and AKAZE only find 10 and 61 matches. SIFT finds an inlier ratio of 0.3292 while ORB and AKAZE are much higher at 0.5 and 0.607. SIFT was significantly slower though as it took about 1.35 seconds while ORB took about 0.15 seconds and AKAZE took about 0.7 seconds. The ORB warp is shown on the left and the AKAZE on the right. These warps are slightly worse than the SIFT warp shown above as the stitching is not as clean and there is blur and misalignment right on the divide.





In this interpolation comparison we can see that bilinear interpolation on the right produces much more blur than nearest neighbors on the left. This is especially apparent in words like USA, Canada, International Friendly, and TNT WEST. This makes sense since bilinear interpolation takes a weighted average of neighboring pixels for smoother edges, while nearest neighbors copies exact color of the nearest neighbors which creates sharper edges.

